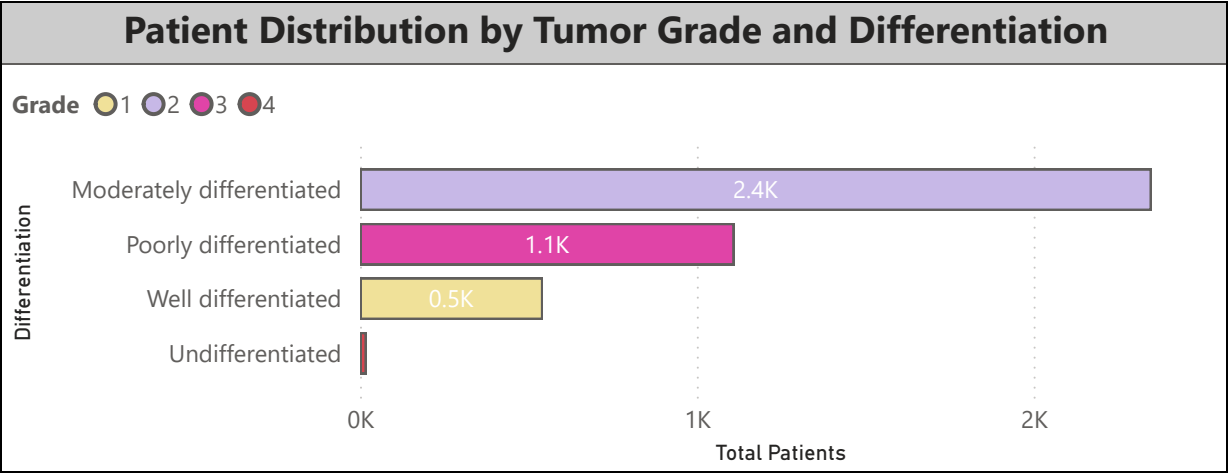
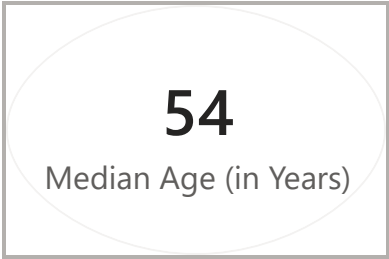
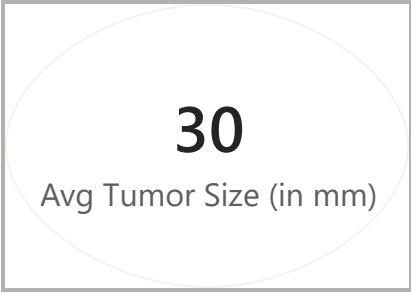
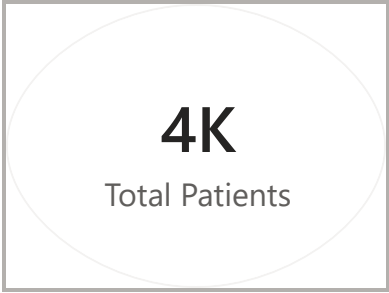
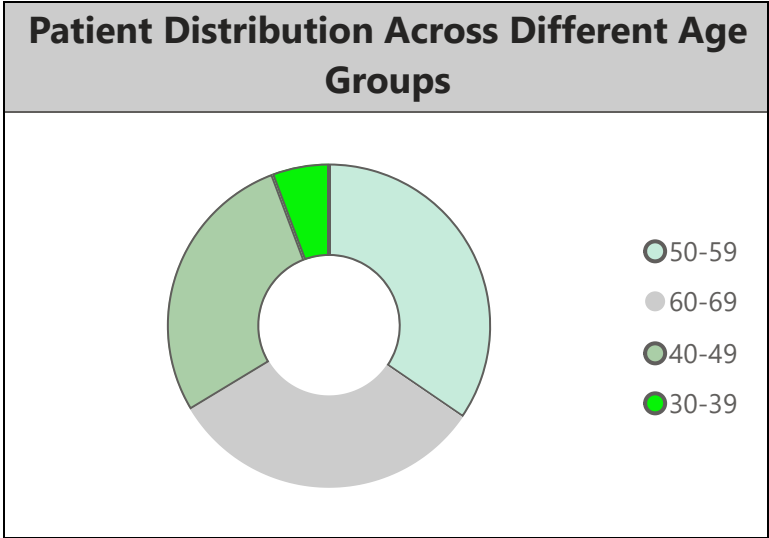
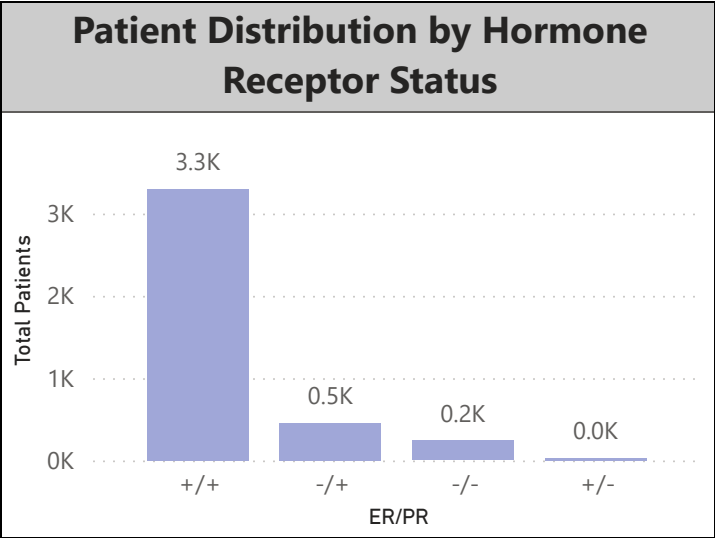


Patient Distribution Across Various Breast Cancer Characteristics

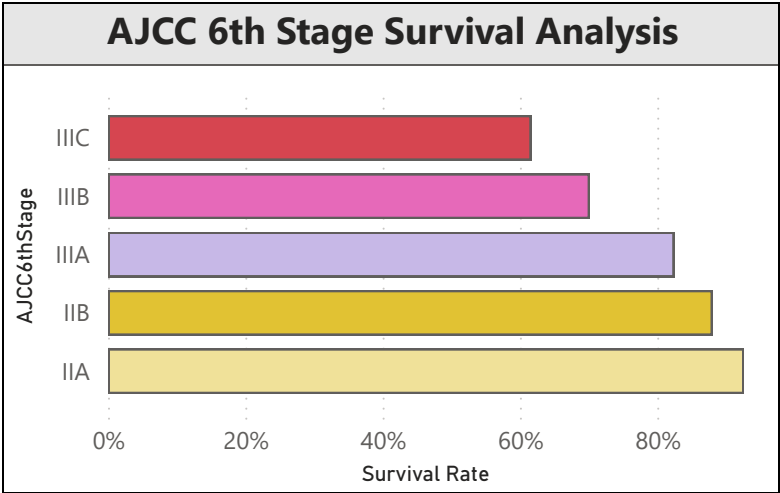
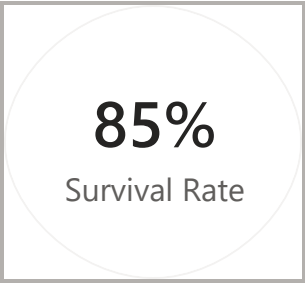
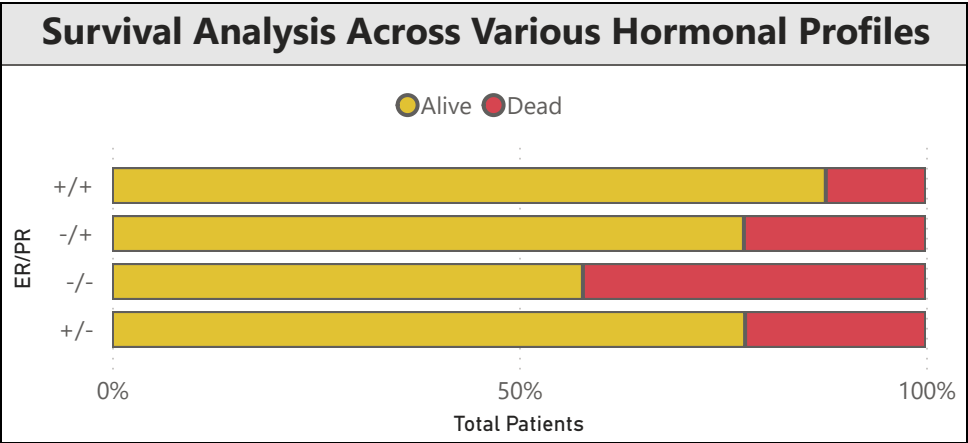
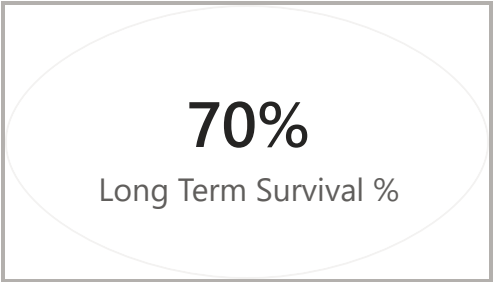
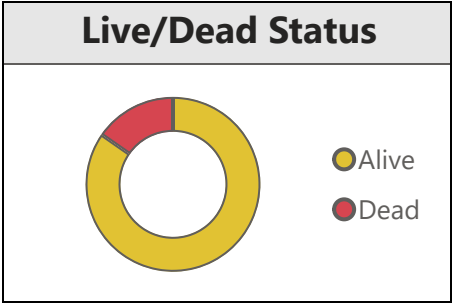
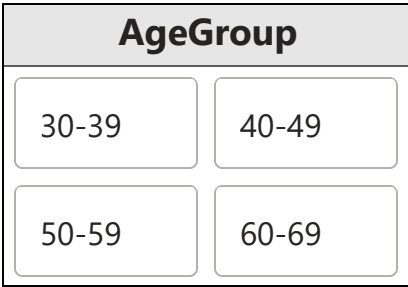
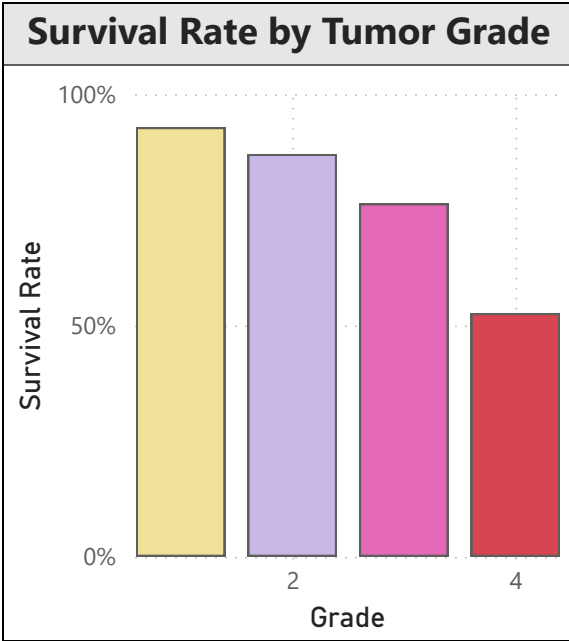


Insights

- Most patients were aged 40 years and above indicating that the risk of breast cancer incidence increased with age.
- A significant proportion were ER+/PR+ which is a good sign since such tumors respond well to hormonal therapy.
- Most of the patients in this cohort were diagnosed with Grade 2 tumors exhibiting moderate differentiation.

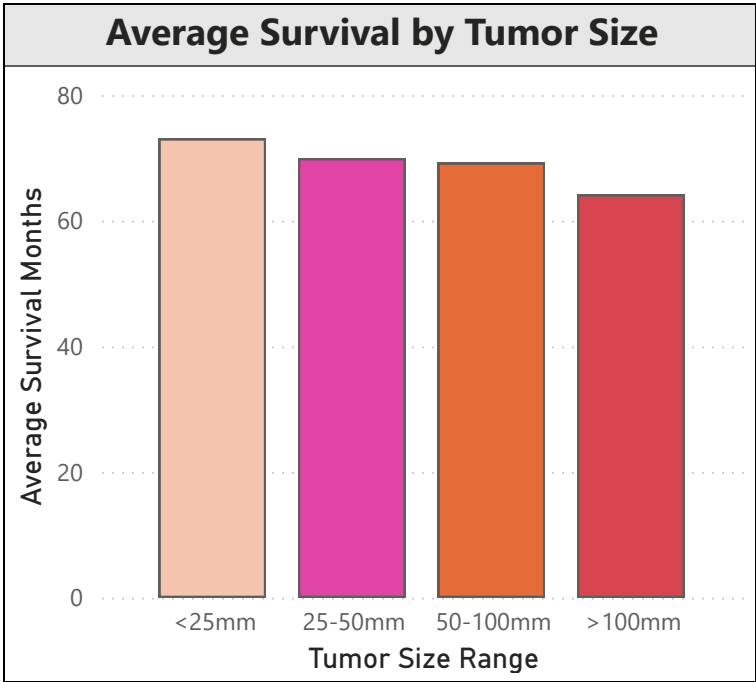
Data Source: Reihan Enamdari. *Breast Cancer Dataset*. Kaggle. Available at: <https://www.kaggle.com/datasets/reihanenamdari/breast-cancer> (Accessed: 20 July 2026).
Derived from the SEER Program of the U.S. National Cancer Institute.

Relationship b/w Survival and Other Variables



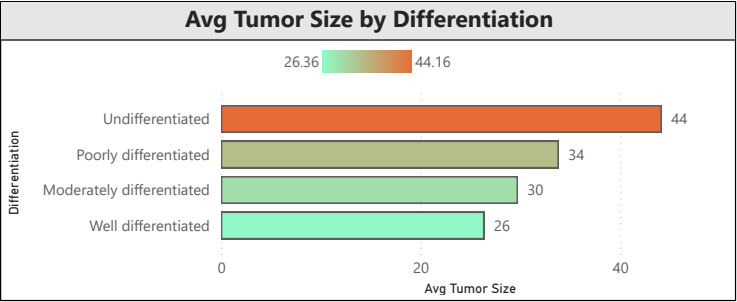
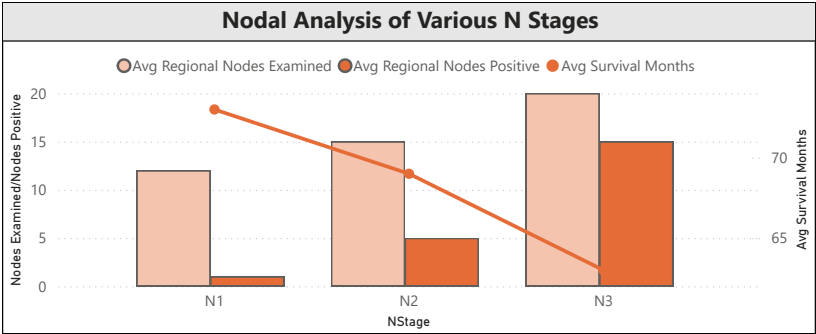
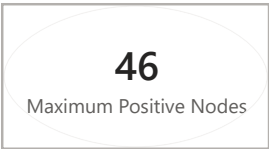
Insights

- Approximately 85% patients were alive by the end of recorded follow-up period.
- The grade and stage of tumor at the time of diagnosis played an important role in survival since the latter drastically reduced at the advanced stage and grade 4.
- The ER-/PR- patients also had a very poor survival outcome.



Clinical Analysis of Tumor Progression

Grade and AJCC 6th Stage Analysis					
Grade	IIA	IIB	IIIA	IIIB	IIIC
1	74.00	71.00	72.00	78.00	63.00
2	73.00	73.00	70.00	69.00	67.00
3	75.00	70.00	68.00	65.00	59.00
4	94.00	49.00	69.00		38.00
Note: Grade 4, IIA (PatientCount = 4); Small sample size, interpret with caution					



Insights

- The largest tumors of the cohort had more undifferentiated cells.
- The average survival of patients at lower grades and stages of tumors was close to average while there was a gradual decline with the tumor progression.
- The gap between number of regional nodes examined and those found to be positive kept widening with progression of tumor stage from N1 to N3, with a decreasing trend in average survival months.